

AI Engineer

Step 6: Deploy AI Models

CareerCompass • Detailed Concept Notes • Research-backed learning guide

Concept & Learning Objective

Training and serving are different workloads. Training may be expensive and iterative; inference should be predictable, observable and fast enough for its users.

A deployment should load a versioned model plus the exact preprocessing needed for inference. The service should validate inputs, enforce resource limits, return predictable errors, expose health information and record safe operational metadata.

Core Concepts — Detailed Breakdown

- Artifact: model weights plus preprocessing/configuration needed for inference.
- Inference API: request validation → preprocessing → model → postprocessing → response.
- Latency/throughput: measure p50/p95/p99 where useful, not only averages.
- Resource management: CPU/GPU, memory, batch size and concurrency.
- Containerization: package runtime dependencies reproducibly.
- Security: authentication/authorization where needed, input limits and no sensitive logging.
- Model monitoring: latency, errors, input distribution drift and model quality where labels become available.
- Versioning: model version, code version and data/preprocessing version should be traceable.
- Rollback: keep a known-good model artifact and deployment procedure.

What You Should Be Able To Do

- Export/load a model.
- Build an inference endpoint.
- Containerize it.
- Add validation and health checks.
- Measure latency.
- Log model version without logging sensitive payloads.

Hands-on Project / Practice

- Deploy a small scikit-learn or TensorFlow model behind a Flask/FastAPI endpoint. Add Docker, health endpoint, input validation, versioned model artifact and a basic monitoring plan.

Recommended References

- Scikit-learn User Guide — https://scikit-learn.org/stable/user_guide.html
- TensorFlow Learn — <https://www.tensorflow.org/learn>
- Python Documentation — <https://docs.python.org/3/>

Note: These notes are designed to teach the concepts and provide a practical learning path. Always prefer the linked official documentation for current APIs, SDK versions and platform-specific release requirements.